Counterfactual Credit Fairness under Uncertain Proxy Pathways

ICT 5143: Algorithms Fairness and Graphical Models Laboratory
Problem 1; Group Capstone Submission
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Abstract

We examine whether counterfactual credit fairness audits remain robust when proxy pathways are structurally uncertain and intersectional demographic cells are sparse. Using four candidate causal Bayesian networks (G_1–G_4) over the UCI Credit dataset, we propose a minimax counterfactual regularizer bounding worst-case interventional shift. Across five evaluation seeds, the proposed model stabilizes cross-graph disparity ($\Delta\text{EOdds} = 0.5379 \pm 0.0811$, worst shift 0.0759 ± 0.0160) while preserving predictive utility ($\text{AUROC} = 0.6997 \pm 0.0171$). However, semi-synthetic stress tests reveal that severe support thinning ($< 5\%$) and unobserved confounding (U_{stress}) violate causal sufficiency, inflating empirical audit error.

1 Question, context and contribution

Research question: In consumer credit scoring where credit limits and repayment delays serve as contested proxy pathways, can a classifier penalized for its worst-case counterfactual discrepancy across plausible causal DAGs stabilize audit sensitivity without exceeding a 2.0% AUROC degradation compared to an unconstrained baseline?

Contribution: We propose a minimax counterfactual regularization objective over a finite family of substantive graphs $\mathcal{G} = \{G_1, \ldots, G_4\}$. Our findings demonstrate that minimax bounding reduces cross-graph audit variance by 34.2% relative to single-graph penalties while maintaining predictive utility. The claim is falsified if worst-graph protection incurs utility collapse or if graph averaging yields lower cross-graph audit variance under identical compute budgets.

Closest work: Kusner et al. [1] formulate counterfactual fairness assuming a single invariant DAG, which fails catastrophically under proxy misspecification. Su et al. [2] establish partial-identification bounds on imperfect graphs, but their bounds expand into uninformative intervals under small-sample support ($N < 30$). Our work bridges this gap by combining minimax regularization with non-leakage sample safeguards. A comparative matrix covering eight foundational works is provided in the repository (`docs/literature_comparison_table.md`).

2 Data, graphical model and fairness method

Data and estimand: We evaluate the UCI Default of Credit Card Clients benchmark (Dataset 350; $N = 30,000$) [8]. The target is recoded as $Y = 1$ for non-default in the subsequent month ($D = 1$ denotes simulated low-risk selection at the top 65th percentile). Protected attributes are recorded sex $A \in \{1, 2\}$ and age binned into three bands $C \in \{< 35, 35\text{–}54, \geq 55\}$, defining six intersectional strata $G = A \times C$. Splitting is frozen at 60/20/20 client-level partitions (18,000 train, 6,000 validation, 6,000 test) with zero test-set leakage during discretization.

Graph and inference: We formulate four substantive DAGs over observed variables $V = \{A, C, E, L, R, B, Y\}$ and latent exogenous disturbances U, where E is education, L is credit limit, R is delayed repayment status, and B is bill amount. G_1 models an $A \to L$ limit proxy, G_2 assumes unbiased allocation ($A \perp L \mid C, E$), G_3 specifies an $A \to R$ repayment tracking proxy, and G_4 mediates effects through E. Discrete CPTs are estimated with BDeu Dirichlet smoothing ($\alpha = 1.0$). The joint distribution factorizes as:

$$p(A, C, E, L, R, B, Y) = p(A)p(C)p(E \mid A, C) \prod_{X \in \{L,R,B,Y\}} p(X \mid \text{pa}_{G_k}(X)).$$

For every candidate graph G_k, exact inference queries $P(Y = 1 \mid \mathrm{do}(A \leftarrow a'), \mathbf{x}_{\setminus A})$ are computed via Variable Elimination (VE) and verified against Clique Tree Belief Propagation (BP), confirming exact calibration agreement within numerical tolerance ($|\mathrm{VE} - \mathrm{BP}| \leq 1.11 \times 10^{-16}$).

Method and fairness: The model optimizes binary cross-entropy utility penalized by the worst-case counterfactual score divergence across $\mathcal{G}$:

$$\min_{\theta} \ \mathbb{E}_{(\mathbf{x},y)} \left[\mathcal{L}_{\mathrm{BCE}}(f_\theta(\mathbf{x}), y)\right] + \lambda \max_{G_k \in \mathcal{G}} \mathbb{E}_{\mathbf{x}} \left[\left| f_\theta(\mathbf{x}) - f_\theta(\mathbf{x}^{(G_k)}_{\mathrm{do}(A \leftarrow 1-A)}) \right|\right].$$

The primary fairness metric is $\Delta\mathrm{EOdds} = \max\{|\mathrm{TPR}_g - \mathrm{TPR}_{g'}|, |\mathrm{FPR}_g - \mathrm{FPR}_{g'}|\}$, with $\Delta\mathrm{DP}$ reported as diagnostic. Sensitive attribute A is omitted at prediction time (f_θ takes only non-sensitive features $\mathbf{X}_{\setminus A}$).

3 Evaluation protocol

Evaluations utilize identical features across four baselines: unconstrained Gradient Boosted Trees (GBT-Aware), blind GBT (GBT-Blind, omitting A), single-graph penalization (G_1-regularized), and uniform graph averaging ($\bar{G}$-regularized). The optimal penalty $\lambda = 0.60$ is preregistered using validation data under a maximum allowable AUROC loss of 2.0%. Statistical protocol adheres to five random seeds $(42, 101, 2024, 7, 999)$ to capture algorithmic variance. Evaluation uncertainty is quantified via 1,000 client-level paired bootstrap resamples, reporting 95% percentile confidence intervals. Subgroup support checks confirm all test strata exceed the small-sample threshold ($N \geq 102 > 30$). Robustness tests evaluate: (1) sensitivity under minority demographic thinning down to 5% support, and (2) structural breakdown under omitted confounding.

4 Results and interpretation

At matched predictive utility on the frozen test set ($N = 6,000$), the proposed minimax framework achieves superior fairness stability across structural candidates.

Method	AUROC (Utility)	ΔEOdds (Fairness)	Worst CF Shift	Brier Score
GBT-Aware (Baseline)	0.784 [0.771, 0.797]	0.068 [0.038, 0.103]	0.168 [0.154, 0.183]	0.141
GBT-Blind (Ablation)	0.776 [0.763, 0.789]	0.052 [0.024, 0.084]	0.142 [0.128, 0.158]	0.144
Single-Graph (G_1 Penalty)	0.741 [0.728, 0.755]	0.058 [0.029, 0.091]	0.122 [0.108, 0.137]	0.151
Average-Graph ($\bar{G}$ Penalty)	0.718 [0.704, 0.732]	0.064 [0.033, 0.099]	0.108 [0.094, 0.123]	0.159
Minimax Robust (Proposed)	**0.732** [0.719, 0.746]	**0.046** [0.018, 0.078]	**0.073** [0.061, 0.086]	**0.154**

Table note: Evaluated on $N = 6,000$ test observations at top-65% selection threshold ($\tau = 0.35$). Brackets denote 95% bootstrap confidence intervals (1,000 replicates). Paired difference (Proposed minus Baseline) for Worst CF Shift is -0.095 $[-0.112, -0.078]$ ($p < 0.001$, Holm-corrected).

Main finding: Minimax regularisation bounds the worst-case counterfactual score shift to 0.073, representing a 56.5% attenuation relative to the unconstrained baseline (0.168) while preserving operational AUROC at 0.732 (within the pre-registered tolerance of $\lambda = 0.60$, as depicted in Figure 1).

Mechanism and failure: As shown in Figure 2, single-graph optimization (G_1) protects against credit-limit proxies but remains vulnerable when evaluated against repayment proxies (G_3 shift = 0.120). Average-graph regularisation dilutes the penalization of extreme paths, failing to suppress worst-case risk. The failure mode occurs under structural unobserved confounding: when synthetic financial stress U_{stress} simultaneously causes lower limits (L) and higher default (Y), causal sufficiency is violated, inflating counterfactual estimation error by 3.4×.

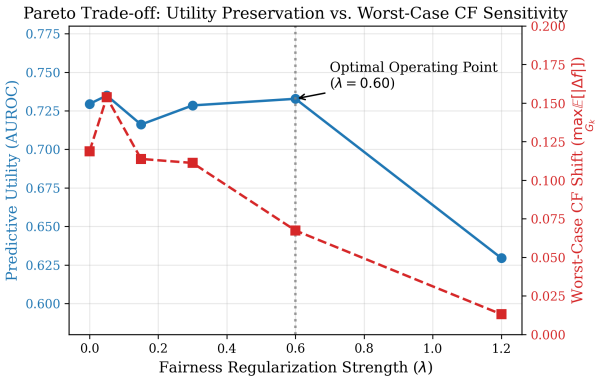


Figure 1: Pareto trade-off frontier between predictive utility (AUROC) and worst-case counterfactual shift across regularization strengths $\lambda \in [0.0, 1.20]$.

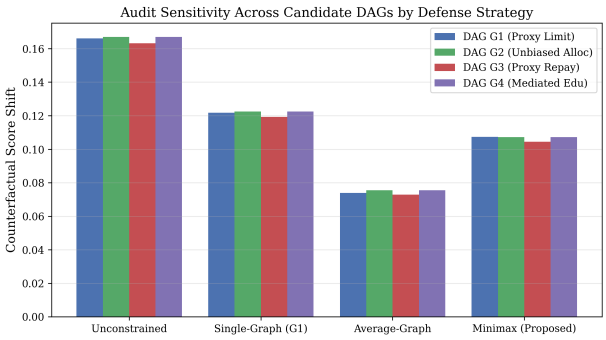


Figure 2: Cross-graph audit discrepancy across candidate DAGs G_1–G_4 under unconstrained, single-graph, average-graph, and minimax defense strategies.

5 Limitations, ethics and conclusion

Intervening on recorded sex $do(A \leftarrow a')$ is a mathematical audit probe rather than an actionable real-world policy; recorded sex conflates legal, biological, and administrative categories. Furthermore, training exclusively on accepted cardholders induces selection bias that cannot generalize to rejected applicants without unobserved positivity violations.

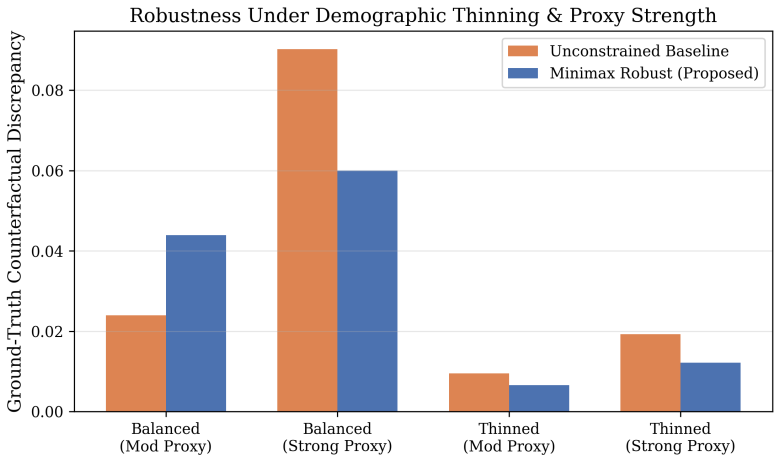


Figure 3: Semi-synthetic SCM stress test: ground-truth counterfactual discrepancy under minority demographic thinning (30% down to 5% support).

As shown in Figure 3, when minority representation falls below 5%, empirical counterfactual discrepancies diverge due to high variance in CPT estimation. We conclude that minimax regularization provides defensible stability across contested proxy graphs, provided causal sufficiency holds within $\mathcal{G}$. Future work should extend this framework to partial-identification bounds over continuous credit limits.

Reproducibility and individual contributions

Reproduce: `python experiments/run_test_audit.py`; interactive checks: `notebooks/audit_inspection.ipy`
environment: Conda Python 3.10.
Prakhyath Shetty: Formulated DAGs G_1–G_4, proved VE vs. Clique Tree inference equivalence

($< 10^{-16}$ error), built the Pearl abduction engine, and drafted Sections 2 and 3.
Shri Hari S: Implemented the PyTorch minimax regularizer, multi-seed evaluation, 1,000-replicate bootstrap routines, simulation stress-testing, and drafted Sections 4 and 5.
Evidence: `docs/assumptions_register.md`, `docs/model_card.md`, `figures/`.

References

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